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Project 2: Exploratory Data Analysis (EDA)

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on an e-commerce transaction dataset to identify trends, patterns, distributions, and outliers while generating meaningful business insights.

Dataset Overview

Metric	Value
Total Records	1200
Total Columns	14
Missing Values	309
Duplicate Records	0
Dataset Type	E-commerce Transactions

Columns:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

Data Quality Analysis

Missing Values

Only one column contained missing values:

Column	Missing Values
Coupon Code	309

Observation:

The missing Coupon Code values indicate that many customers completed purchases without using promotional discounts.

Duplicate Records

Duplicate Rows Found: 0

Observation:

No duplicate records were found, indicating strong data consistency.

Descriptive Statistics

Quantity

Statistic	Value
Mean	2.95
Median	3
Minimum	1
Maximum	5

Observation

Most customers purchase approximately 3 items per transaction.

Unit Price

Statistic	Value
Mean	356.41
Median	364.21
Minimum	11.39
Maximum	699.93

Observation

Products cover a wide pricing range, indicating a mix of low-cost and premium items.

Total Price

Statistic	Value
Mean	1053.97
Median	823.62
Minimum	11.39
Maximum	3456.40

Observation

The average transaction value exceeds 1000, showing relatively high-value purchases.

Items In Cart

Statistic	Value
Mean	5.49
Median	5
Minimum	1
Maximum	10

Observation

Most customers add around 5 products to their shopping cart.

Product Analysis

Most Frequently Ordered Products

Product	Orders
Printer	181
Tablet	179
Chair	178
Laptop	173
Desk	170

Observation

Printer was the most frequently purchased product.

Highest Revenue Generating Products

Product	Revenue
Chair	195,620.11
Printer	195,612.61
Laptop	192,126.56
Tablet	186,568.95
Monitor	175,651.41

Observation

Although Printer received the highest number of orders, Chair generated the highest overall revenue.

Payment Method Analysis

Payment Method	Orders
Online	258
Cash	246
Credit Card	234
Debit Card	232
Gift Card	230

Observation

Online payment is the most preferred payment method among customers.

Order Status Analysis

Status	Count
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

Observation

Cancelled and Returned orders represent a significant portion of transactions and may indicate customer satisfaction or operational issues.

Outlier Detection

Outlier analysis was performed on the TotalPrice column using the Interquartile Range (IQR) method.

Result

Outliers Detected: 8

Observation

Only 8 transactions were identified as outliers, indicating that the dataset is generally stable.

Possible reasons:

- Bulk purchases
- Premium products
- Business customers placing large orders

Key Insights

Insight 1

Average order value is:

1053.97

indicating strong customer spending.

Insight 2

Printer is the most ordered product.

Insight 3

Chair generates the highest revenue.

Insight 4

Online payment is the preferred payment method.

Insight 5

The dataset contains very few outliers, demonstrating consistent customer purchasing behavior.

Conclusion

The Exploratory Data Analysis revealed valuable business insights regarding customer purchasing behavior, product performance, payment preferences, and transaction patterns. The dataset exhibited high data quality with no duplicate records and only one column containing missing values. Revenue analysis identified Chair and Printer as the strongest-performing products, while payment analysis showed a clear preference for online transactions. Overall, the dataset is reliable and suitable for advanced analytics, dashboard creation, and future predictive modeling.